Meaning as Direction in Embedding Space

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Where this fits: Pipeline map step 5 (Engineer features). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	**5 Features**	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Dot product; Word embeddings.
- **Leads to:** Contextual embeddings; Superposition and nearly perpendicular directions.

1. Overview

Key point: In a trained **word embedding** (G-2127), meaning lives in **directions** (G-613): the arrow from "man" to "woman" points roughly the same way as the arrow from "uncle" to "aunt", so adding it to "uncle" lands next to "aunt". A **dot product** (G-634) with a direction measures how much of that meaning a word carries, in GloVe and in GPT-2's first layer alike.

A **word embedding** (G-2127) gives each word a dense vector (a short list of numbers, for example 100), learned so that words used in similar ways get similar vectors. The first look at embeddings used *distances*: similar words sit close together. This Note looks at *directions*: what the arrow between two word vectors means, and what we can do with it.

We use two real embedding tables:

1. **GloVe** (G-851), 100 numbers per word, trained on 6 billion **tokens** (G-1981; here single words and punctuation marks) of Wikipedia and news text (Pennington et al. 2014). We keep its 100,000 most frequent words.
2. **GPT-2 small's token-embedding table** (G-1980), 768 numbers per token (for GPT-2, a word or a piece of a word), the first layer of a real transformer (Radford et al. 2019).

Figure 1 shows the central idea on real GloVe vectors. We cannot draw 100 dimensions (each of a vector's 100 numbers is one dimension), so the figure draws the flat sheet (a plane) that passes through the points for man, woman and king. Watch the orange arrow "woman − man" being lifted onto "king": its tip lands near "queen", but not on it.

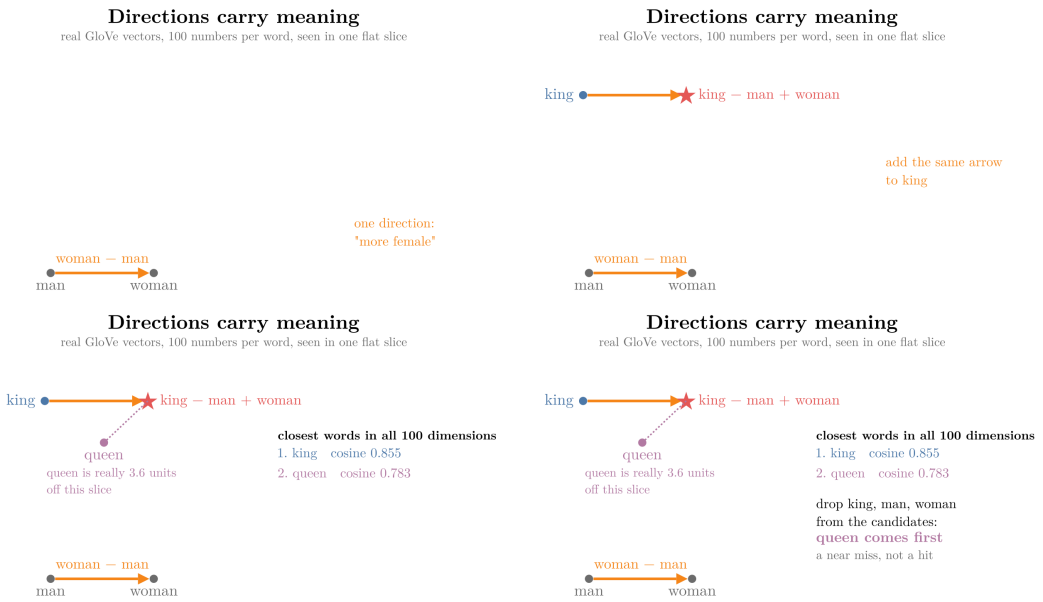


Figure 1: King − man + woman on real GloVe vectors, drawn in the flat slice through man, woman and king. The woman − man arrow, moved to king, ends at the star. In all 100 dimensions the nearest word to the star is still "king" (cosine 0.855); "queen" is second (0.783) and comes first once king, man and woman are removed from the candidates

> **Extra:** Two ideas of this Note follow Sanderson's "Transformers, the tech behind LLMs" (3Blue1Brown, Deep Learning Chapter 5, 2024): adding the woman − man difference to king and searching for the nearest word, which he notes lands only "kind of" near queen, with family relations working better (14:37–15:39); and taking the dot product of cats − cat with singular and plural nouns and with one, two, three, ... as a plurality **probe** (G-1575) (16:44–17:44). The figures, their design and their data are our own.

2. Prerequisites

- **Word embeddings** (G-2127): dense, learned vectors; the embedding layer as a lookup table.
- **Cosine similarity** (G-491): the **dot product** (G-634), its sign, and cosine similarity as the angle between two vectors.
- The dot product as a projection: the dot product with a **unit vector** (G-2048) is the length of the projection onto it.
- Three views of a vector: a vector as a point or an arrow in n dimensions; subtracting two vectors gives the arrow from one to the other.

3. A word is a point; nearby points share meaning

> **Key point:** Each word is one point in a 100-dimensional space (a vector of 100 numbers, one number per dimension). Words whose vectors make a small angle (high cosine similarity) are used in similar ways.

GloVe (G-851) gives each word a vector of 100 numbers. We cannot draw 100 dimensions, but we can measure angles. The **cosine similarity** (G-491) of two vectors (see the definition and range) is 1 when they point the same way and 0 when they are perpendicular. The 8 words with the highest cosine similarity to "tower" are (Notebook): towers (0.85), building (0.73), dome (0.69), spire (0.68), gate (0.67), skyscraper (0.67), roof (0.66) and walls (0.66) (Figure 2).

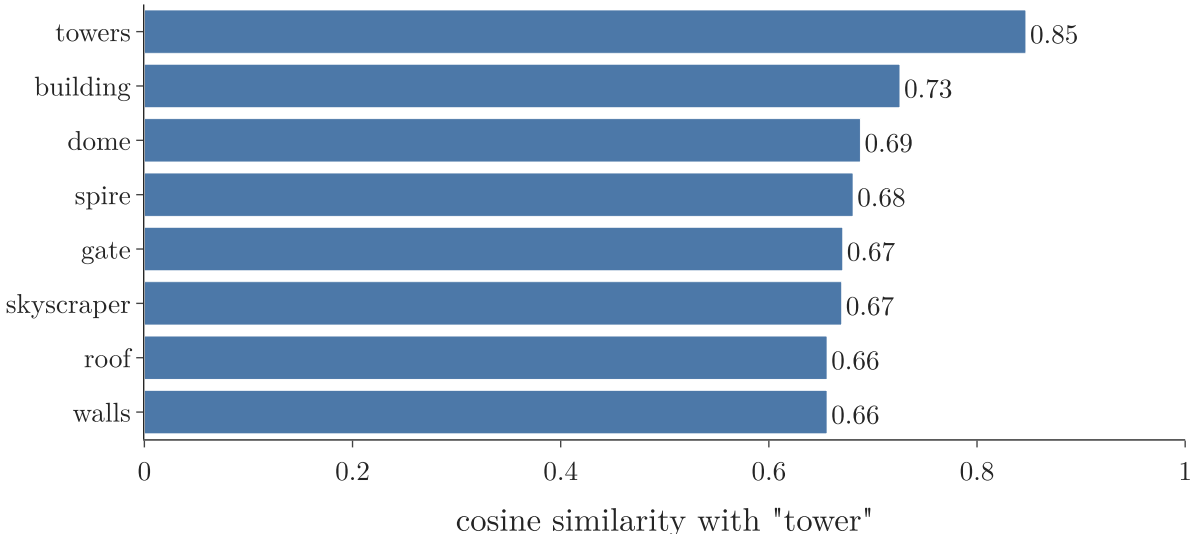


Figure 2: The 8 GloVe words closest to "tower" by cosine similarity. All are tall structures or parts of buildings

All of them are tall structures or parts of buildings. Distances say *which* words are alike. They do not yet say *how* two words differ. For that we look at the arrow between them.

4. The difference between two words is a direction

> **Key point:** The arrow from a male word to its female counterpart points roughly the same way for many pairs. In **GloVe** (G-851) its cosine with woman − man is between 0.44 and 0.67 for 11 of 12 pairs, while 99 percent of random word pairs stay within ±0.30.

Subtracting two vectors gives the arrow from one point to the other. Call $g = e_{\text{woman}} - e_{\text{man}}$ the "woman − man" arrow. If the embedding stores gender as a **direction** (G-613), then $e_{\text{aunt}} - e_{\text{uncle}}$, $e_{\text{mother}} - e_{\text{father}}$ and so on should point roughly along g.

We test this with the cosine between each pair's arrow and g (Figure 3). As a baseline, the Notebook draws 5,000 random pairs of common words and measures the same cosine for their arrows.

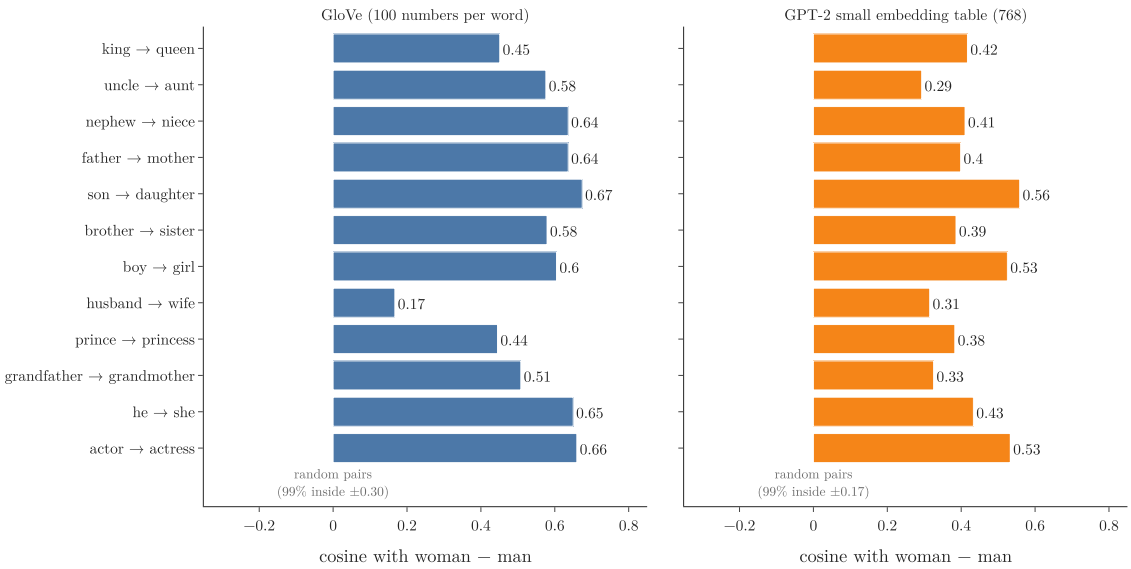


Figure 3: Cosine between each pair's arrow (female word − male word) and woman − man. Grey band: 99 percent of random word pairs fall inside it. Left: GloVe. Right: GPT-2 small's embedding table

- **GloVe:** 11 of the 12 pairs have cosines from 0.44 (prince → princess) to 0.67 (son → daughter). Random pairs average 0.00, and 99 percent lie within ±0.30. The exception is husband → wife, at 0.17.
- **GPT-2:** all 12 pairs lie between 0.29 and 0.56, above its random band of ±0.17.

The arrows are far from identical: a cosine of 0.6 is an angle of about 53 degrees. But they all lean the same way, much more than chance allows. Somewhere among the 100 directions, one direction tracks "female versus male".

Why training produces directions. GloVe was designed for exactly this. Its authors start from co-occurrence counts: how often word k appears near word i, giving a **co-occurrence probability** (G-404) P_{ik}. Their Table 1 shows that a *ratio* of two such probabilities picks out what distinguishes two words: "solid" appears 8.9 times more often near "ice" than near "steam", "gas" about 12 times less often, and "water", related to both, about equally often (ratio 1.36). GloVe trains the vectors so that a word vector w_i dotted with a context vector $\tilde{w}_k$ gives the log of the probability (Pennington et al. 2014, §3, eqs. 1–7). For two words i and j and the same context word k:

$$w_i \cdot \tilde{w}_k \approx \log P_{ik}$$

$$w_j \cdot \tilde{w}_k \approx \log P_{jk}$$

Subtract the second line from the first, and use the rule $\log a - \log b = \log(a/b)$:

$$(w_i - w_j) \cdot \tilde{w}_k \approx \log P_{ik} - \log P_{jk}$$
$$(w_i - w_j) \cdot \tilde{w}_k \approx \log \frac{P_{ik}}{P_{jk}}$$

For i = ice, j = steam and k = solid, the ratio is 8.9, so the right side is (natural log):

$$\log 8.9 = 2.19$$

The arrow from steam to ice has a large **dot product** (G-634) with the context vector of "solid".

So the arrow between two words is trained to encode how their contexts differ. "Uncle" and "aunt" differ in their contexts much as "man" and "woman" do (for example, in how often "he" or "she" appears nearby), so their arrows end up pointing in similar directions. GPT-2 learns its table differently, by predicting the next token, yet Figure 3 shows the same structure.

5. Arithmetic with directions

> **Key point:** Adding the woman − man arrow to "uncle", "nephew" or "father" lands closest to "aunt", "niece" and "mother". For "king" the result is a near miss: "king" itself stays the closest word, and "queen" is second.

If g means "more female", then adding g to "king" should land near "queen":

$$e_{\text{king}} + g$$
$$= e_{\text{king}} - e_{\text{man}} + e_{\text{woman}}$$

Pennington et al. (2014, §4.1) answer such an **analogy question** (G-196), "a is to b as c is to ?", with the word whose vector has the largest **cosine similarity** (G-491) with $w_b - w_a + w_c$. Figure 1 animates this rule on the real **GloVe** (G-851) vectors, and the table gives the results (Notebook). "Rank" is the place of the expected word when the three question words are removed from the candidates. Figure 4 draws the same results.

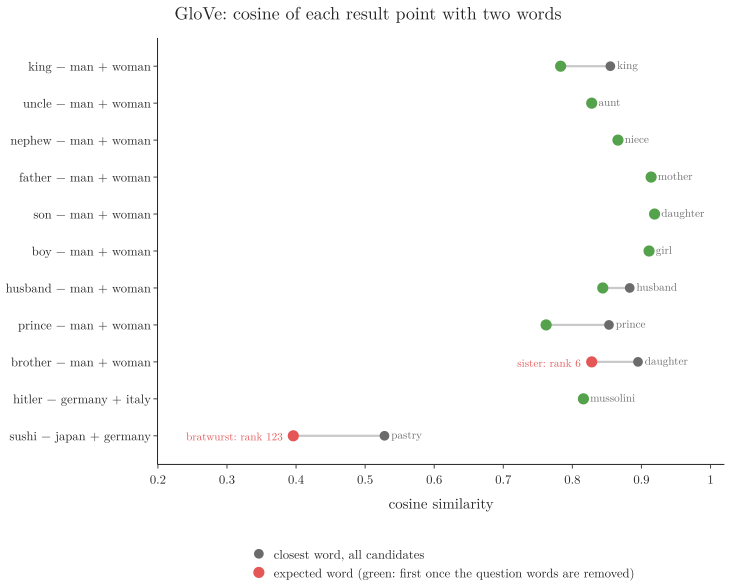


Figure 4: Each analogy question in GloVe: the cosine of the result point with the closest word among all candidates (grey) and with the expected word (green when it ranks first once the question words are removed, red with its rank otherwise). A single dot means the expected word is the closest of all

Question	Closest word, all candidates	Closest without the question words	Rank of the expected word	Its cosine
king − man + woman	**king** (0.855)	queen, monarch, throne	1 (queen)	0.783
uncle − man + woman	aunt	aunt, daughter, mother	1	0.828
nephew − man + woman	niece	niece, daughter, granddaughter	1	0.866
father − man + woman	mother	mother, daughter, wife	1	0.914
son − man + woman	daughter	daughter, wife, mother	1	0.919
boy − man + woman	girl	girl, mother, child	1	0.911
prince − man + woman	**prince**	princess, daughter, queen	1	0.762
husband − man + woman	**husband**	wife, mother, daughter	1	0.844
brother − man + woman	daughter	daughter, wife, mother	**6** (sister)	0.828

Question	Closest word, all candidates	Closest without the question words	Rank of the expected word	Its cosine
hitler − germany + italy	mussolini	mussolini, fascist, stalin	1	0.816
sushi − japan + germany	pastry	pastry, ducasse, gourmet	**123** (bratwurst)	0.396

Three things stand out.

1. **Family words work best.** For uncle, nephew, father, son and boy the expected word is the closest word of all 100,000, even with the question words left in.
2. **King → queen is a near miss.** With every word allowed, the closest word to king − man + woman is "king" itself (cosine 0.855); "queen" is second (0.783). By plain distance the point is also closer to king (3.36) than to queen (4.08). In Figure 1's flat slice, queen appears close by, but its true position is 3.6 units off that slice. The arrow moves "king" *towards* "queen" without reaching it.
3. **Not every pair works.** brother − man + woman ranks "sister" only sixth, behind daughter, wife, mother, husband and father. And in GloVe, sushi − japan + germany gives "pastry"; "bratwurst" ranks 123rd.

> **Extra:** The **direction** (G-613) is learned from how words are used, so it carries whatever the text associates with each word. "Queen" is used for monarchs, but also for bands, chess pieces and card games, so its vector is not just "female king". Static embeddings hold an average meaning measures the same averaging effect for "bank".

6. A dot product reads how much of a direction a word has

> **Key point:** Average the arrows from singular to plural over 10 nouns to get a "plural direction". Its **dot product** (G-634) with a word's vector scores how plural the word is: on 12 nouns not used to build it, every plural scores above its singular, in GloVe and in GPT-2.

Section 4 compared two arrows. A **direction** (G-613) can also be used as a **probe** (G-1575) for a single word. Take a **unit vector** (G-2048) p (length 1). By the dot product as a projection, the dot product $e \cdot p$ is the length of the shadow of e on the line through p: large and positive when e points along p, zero when it is perpendicular, negative when it points away.

A tiny case with 2-number vectors: the unit vector is $p = (0.6, 0.8)$. Its length is 1:

$$\sqrt{0.36 + 0.64} = 1$$

A word with $e = (2, 1)$ and another with $e = (-1, 0.5)$ score:

$$e \cdot p = 2 \times 0.6 + 1 \times 0.8 = 2.0$$

$$e \cdot p = (-1) \times 0.6 + 0.5 \times 0.8 = -0.2$$

The first word points along p (large, positive), the second slightly away from it (small, negative). The three steps of the probe:

1. **Build the direction:** average the arrows (plural − singular) of 10 pairs, such as cat → cats, dog → dogs, city → cities, then scale the result to length 1. This is p, the **plural direction** (G-1506).
2. **Score a word:** compute $e_{\text{word}} \cdot p$.
3. **Test on new words:** 12 other pairs, from apple/apples to teacher/teachers, none of them used in step 1.

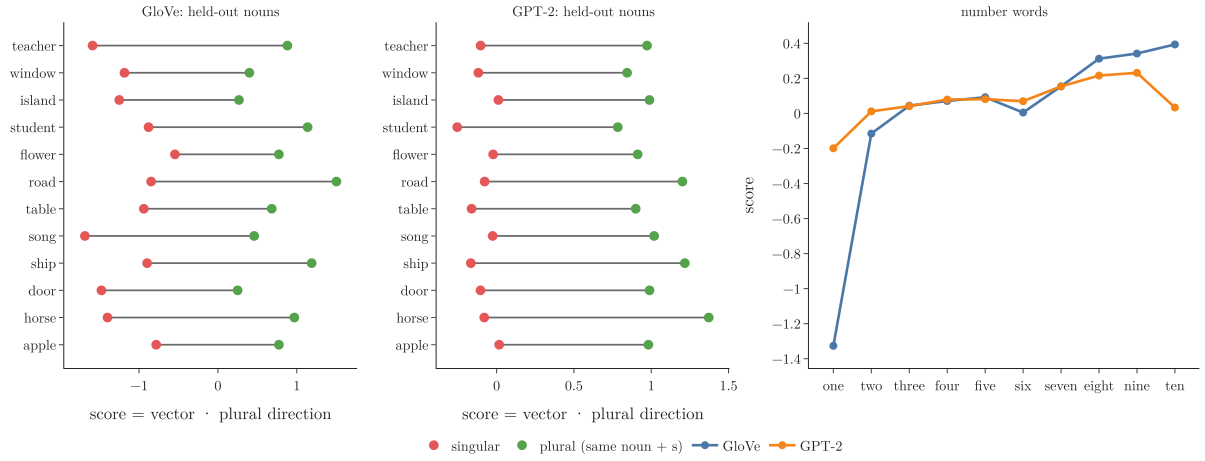


Figure 5: Left and middle: scores of 12 held-out nouns on the plural direction; red the singular, green the plural. Right: scores of the number words one to ten

Figure 5 gives the results (Notebook):

	GloVe	GPT-2 small
Held-out pairs where the plural scores higher	12 of 12	12 of 12
Mean score, singular nouns	−1.12	−0.09
Mean score, plural nouns	0.77	1.02
Rank correlation of the scores of one … ten with 1 … 10 (1 means the scores rise in exactly the order of the numbers)	0.93	0.62

The probe separates singular from plural for every new noun. On the number words the picture is rougher. In **GloVe** (G-851), "one" scores lowest by far (−1.33), and the scores rise, with small dips, up to "ten" (0.39). In GPT-2 they rise from "one" (−0.20) to "nine" (0.23), but "ten" drops back to 0.03. The direction captures "one versus many" clearly and "how many" only roughly.

This is the same dot product that attention uses to compare a query with a key (see dot products as similarity): a large dot product means the two vectors point the same way.

7. The same structure inside a transformer

> **Key point:** GPT-2 small's token-embedding table, the first layer of a real transformer, passes the same tests: all 9 family analogies put the expected word first once the question words are removed, and the plural **probe** (G-1575) scores 12 of 12.

GPT-2 small turns each token into a 768-number vector by looking it up in a table of 50,257 rows (see token and position embeddings). The table is learned while the model learns to predict the next token, not by a co-occurrence objective like **GloVe** (G-851). Every word we test is a single GPT-2 token with a leading space, such as " king".

Question	Closest, all candidates	Closest without the question words	Its cosine
king − man + woman	king	queen	0.709
uncle − man + woman	uncle	aunt	0.696
nephew − man + woman	nephew	niece	0.797
father − man + woman	father	mother	0.753
son − man + woman	daughter	daughter	0.817
boy − man + woman	girl	girl	0.798
husband − man + woman	husband	wife	0.703
prince − man + woman	prince	princess	0.706
brother − man + woman	brother	sister	0.747

Once the three question words are removed, all 9 expected words come first (Figure 6), including "sister", which GloVe ranked sixth. With every word allowed, the starting word itself is the closest in 7 of the 9 questions: the arrow moves the point towards the answer, but in GPT-2's table it usually does not carry it past the starting word.

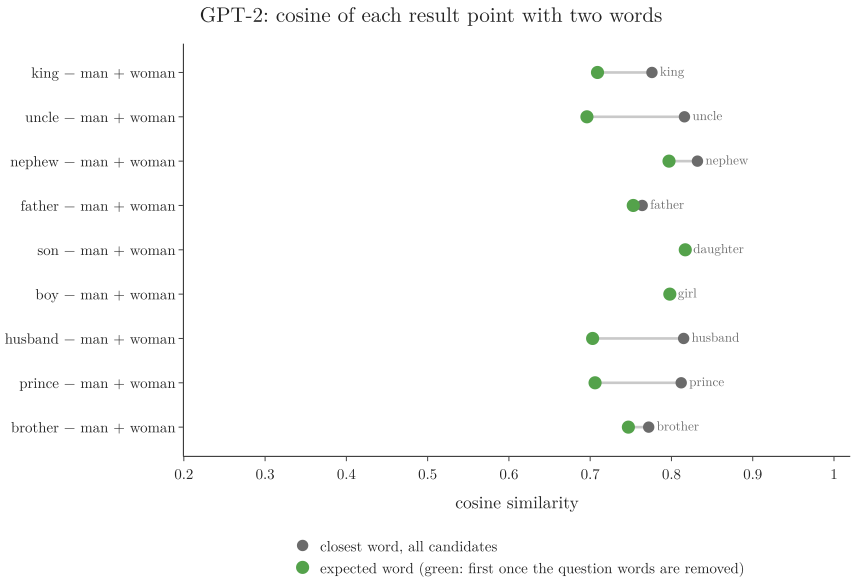


Figure 6: The same analogy chart for GPT-2 small's token-embedding table. Every expected word is first once the question words are removed (all green); in 7 of 9 questions the starting word is still the closest of all

Two links to later Notes:

- These are only the *first* vectors of GPT-2. Its attention layers then change each vector according to the words around it (static and contextual embeddings).
- GPT-2 uses this same table a second time, at the very end, to turn its last vector into a score for every token (see the unembedding). A **direction** (G-613) in this table is therefore also a direction the model can read out.

8. Summary

Idea	What we measured (GloVe)	GPT-2 small table
Nearby points share meaning	the 8 nearest words to "tower" are all structures	(not measured)
A word pair's arrow is a direction	female − male arrows: cosine 0.44–0.67 with woman − man for 11 of 12 pairs; random pairs within ±0.30	0.29–0.56; random within ±0.17
Arithmetic with directions	uncle, nephew, father, son, boy → right word first; king → "queen" second, after "king"; brother → "sister" sixth	all 9 right once the question words are removed
A dot product is a probe	plural direction: 12 of 12 held-out plurals above their singular	12 of 12

- Meaning sits in **directions** (G-613): the arrow between two related words points the same way for many pairs, because pairs such as uncle/aunt and man/woman differ in their contexts in the same way.
- Adding a direction to a word moves it towards the matching word. In **GloVe** (G-851) it lands on the answer for uncle, nephew, father, son and boy; for king, prince and husband the starting word stays closest; brother → sister ranks only sixth. So treat word arithmetic as a push in the right direction, not an exact answer.
- A **dot product** (G-634) with a unit direction measures how much of that meaning a word carries, because it is the length of the word's shadow on that direction; this is the same dot product attention uses to compare a query with a key.
- GloVe is trained so that word differences encode context differences; GPT-2's embedding table, trained only to predict the next token, shows the same structure, so the idea carries over to the first layer of a real transformer.
- So a word vector is a point whose directions carry meaning, and a dot product is the tool that reads that meaning out.

9. Sources

Built from

- Sanderson, G. (3Blue1Brown), "Transformers, the tech behind LLMs | Deep Learning Chapter 5", 2024, 3blue1brown.com/lessons/gpt, https://www.youtube.com/watch?v=wjZofJX0v4M. Directions carry meaning; king plus woman − man lands near queen, only roughly; the cats − cat plurality probe and the number words (13:33–17:44).
- Pennington, J., Socher, R. and Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. *EMNLP 2014*, pp. 1532–1543. aclanthology.org/D14-1162. Abstract and §1 (king − queen = man − woman); §3, Table 1 (ice/steam co-occurrence ratios)

and eqs. 1–7 (vector differences encode log ratios of co-occurrence probabilities); §4.1 (analogy questions answered by the largest cosine with $w_b - w_a + w_c$); §4.2 (the 6-billion-token Wikipedia 2014 + Gigaword 5 corpus, 400,000-word vocabulary).

Other references

- GloVe project page, nlp.stanford.edu/projects/glove (checked 3 October 2026): download `glove.6B.zip` ("Wikipedia 2014 + Gigaword 5 (6B tokens, 400K vocab, uncased, 50d, 100d, 200d, & 300d vectors, 822 MB download)"). The Notebook reads only the 100-d file out of it.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. and Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI. Weights: `openai-community/gpt2` (GPT-2 small: 50,257 tokens, 768 numbers per token), huggingface.co.

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Direction (G-613)	The way an arrow (vector) points, regardless of its length. In a word embedding, meaning lives in directions: the arrow from "man" to "woman" points roughly the same way as the one from "uncle" to "aunt".
GloVe (G-851)	Word vectors learned from how often words appear near each other: training makes the dot product of two words' vectors match the log of their co-occurrence probability (Pennington et al. 2014).
Probe (G-1575)	A fixed direction used to score words: the dot product of a word's vector with it.
Co-occurrence probability (G-404)	How likely word k is to appear near word i in the training text, written P_{ik}; GloVe trains word vectors from these probabilities.
Analogy question (G-196)	"a is to b as c is to ?", answered by the word closest to $c - a + b$.
Plural direction (G-1506)	A direction in embedding space: the average of the arrows from singular to plural nouns, scaled to length 1; a word's dot product with it scores how plural the word is.
Word embedding (G-2127)	A learned real-valued vector for each word; words used in similar ways get nearby vectors.

Term	Meaning
Dot product (G-634)	Multiplying two vectors entry by entry and adding the products, $u^\mathsf{T}x$, which gives one number. It is large when the vectors point the same way, so it measures similarity and gives projections.
Token (G-1981)	One unit of text the model reads or writes; here, one word.
Token embedding matrix W_E (G-1980)	The learned table with one vector per token (50,257 vectors of 768 numbers in GPT-2); in GPT-2 also used as the output matrix.
Cosine similarity (G-491)	A score from -1 to 1 for how closely two vectors point the same way: the cosine of the angle between them. It ignores their lengths, so it compares texts or word embeddings by direction only.
Unit vector (G-2048)	A vector of length 1, used to describe a direction.